Physics
Higher level
Paper 1A and Paper 2
B.4 Thermodynamics

8 November 2026

W4ss Academy practice test 1 | **Topic** B.4 Thermodynamics | **Ref.** W4SS–PHY–B4–01

Student name

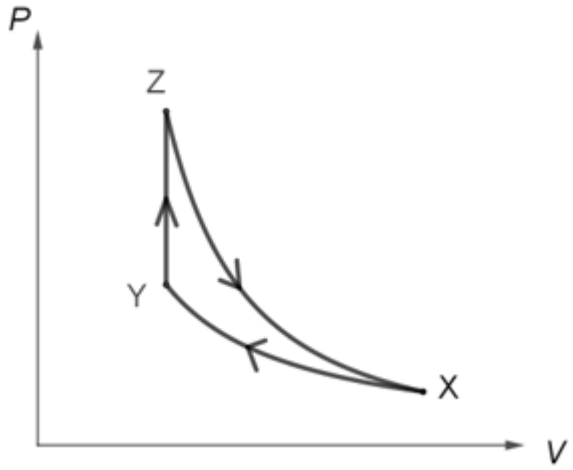
30 minutes [Paper 1A and Paper 2]

Instructions to students

- Write your name in the boxes above.
- Do not open this examination paper until instructed to do so.
- Answer all questions.
- **Paper 1A:** for each question, choose the answer you consider to be the best and indicate your choice on the answer sheet provided.
- **Paper 2:** answers must be written within the answer boxes provided.
- A calculator is required for this paper.
- A clean copy of the **physics data booklet** is required for this paper.
- The maximum mark for paper 1A is **[4 marks]**.
- The maximum mark for paper 2 is **[16 marks]**.
- The maximum mark for this examination paper is **[20 marks]**.

Paper 1A

1. A thermodynamic cycle consisting of an adiabatic, isovolumetric and isothermal processes is shown.



Which of the following correctly identifies the processes of the cycle?

	Adiabatic	**Isovolumetric**	**Isothermal**
A.	X → Y	Y → Z	Z → X
B.	Z → X	X → Y	Y → Z
C.	Z → X	Y → Z	X → Y
D.	Y → Z	Z → X	X → Y

2. An ideal gas expands isothermally. The work done by the gas is 100 J. What is the change in the internal energy of the gas?

A. −100 J

B. 0

C. +50 J

D. +100 J

3. Three statements about the Carnot cycle are:

I. The Carnot cycle is reversible.
II. The net entropy change of the surroundings of the gas over one cycle is positive.
III. Heat transfer takes place in only two stages of the cycle

Which statements are correct?

A. I and II

B. I and III

C. II and III

D. I, II and III

4. An energy of 200 J is transferred isothermally to an ideal gas. The temperature of the gas is 27 °C.

The entropy change of the gas is

A. 0.67 J K^{-1}

B. 0.14 J K^{-1}

C. 1.5 J K^{-1}

D. 7.4 J K^{-1}

Paper 1A — answer sheet

Student name _____ Date _____

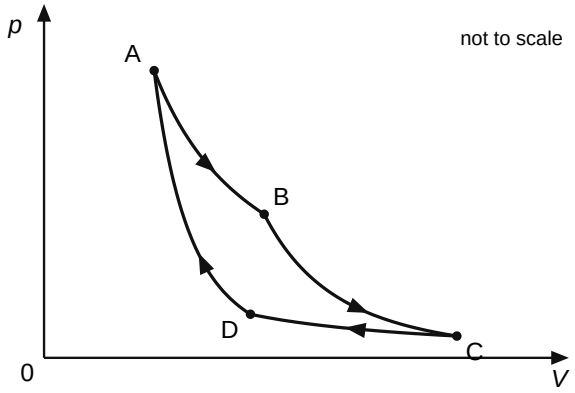
1.	A	B	C	D
2.	A	B	C	D
3.	A	B	C	D
4.	A	B	C	D

Shade **one** circle per question. If you change your answer, cross out the wrong circle clearly.

Paper 2

Answer **all** questions. Answers must be written within the answer boxes provided.

1. An ideal monatomic gas is the working substance of a heat engine. The gas is taken through the Carnot cycle ABCDA shown on the *pV* diagram.



A → B: isothermal expansion at 620 K

B → C: adiabatic expansion

C → D: isothermal compression at 340 K

D → A: adiabatic compression

(a) Show that during an adiabatic expansion of an ideal monatomic gas the temperature *T* and volume *V* are given by

$TV^{\frac{2}{3}}$ = constant [2]

(b) (i) Calculate the efficiency of the cycle. [1]

(Question 1 continued)

(ii) The work done during the isothermal expansion A → B is 540 J. Calculate the thermal energy that leaves the gas during one cycle. [2]

(iii) Calculate the ratio $\frac{V_C}{V_B}$ where V_C is the volume of the gas at C and V_B is the volume at B. [2]

(c) (i) Calculate the change in the entropy of the gas during the change A to B. [1]

(ii) Explain, by reference to the second law of thermodynamics, why a real engine operating between the temperatures of 620 K and 340 K cannot have an efficiency greater than the answer to (b)(i). [2]

2. An insulated container is divided into two equal halves by a partition. A gas is initially confined to the left half (configuration A). The partition is removed and the gas expands to fill the whole container, so that there are molecules in both halves (configuration B).

Treat the molecules as distinguishable. A microstate specifies which half of the container each molecule is in.

(a) Explain why the gas in configuration B has a greater number of microstates than in A. [3]

(b) Deduce, with reference to entropy, that the expansion of the gas from the initial configuration A is irreversible. [3]

Sources of the questions

This practice test was put together by W4ss Academy for internal class use. The questions are taken from the IB DP Physics Questionbank (first assessment 2025), topic B.4 Thermodynamics. The introductions of the Paper 2 questions are missing from the source: W4ss Academy wrote them using only the data of the official markscheme, so that the markscheme still applies. The questions remain the property of the International Baccalaureate Organization and must not be published or sold.

Paper 1A — question 1	EXE.1A.HL.TZ0.17
Paper 1A — question 2	EXE.1A.HL.TZ0.18
Paper 1A — question 3	EXE.1A.HL.TZ0.23
Paper 1A — question 4	EXE.1A.HL.TZ0.21
Paper 2 — question 1	19M.2.HL.TZ1.7 — introduction written by W4ss Academy from the markscheme
Paper 2 — question 2	EXE.2.HL.TZ0.11 — introduction written by W4ss Academy from the markscheme